



Improving PM₁₀ and PM_{2.5} concentration prediction using the Brazilian Regional Atmospheric Modeling 5.2 System in Lima, Peru

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ABSTRACT

Urban air pollution caused by atmospheric aerosols has far-reaching implications on human health, economy, and climate. Using chemical transport models (CTMs) as inputs to the existing vehicle emission inventory (VEI) data in the metropolitan area of Lima and Callao (MALC) results in lower simulated PM₁₀ and PM_{2.5} concentrations than the measured concentrations in MALC. Thus, the model needs to be improved to increase the accuracy of simulations. Numerical air pollution modeling using CTMs has rarely been studied in Peru. This study aimed to improve PM₁₀ and PM_{2.5} concentration simulations using CTMs in Peru by calculating a new VEI. Subsequently, PM₁₀ and PM_{2.5} simulations were conducted using the Brazilian Regional Atmospheric Modeling System (BRAMS) 5.2, which employs both VEI from a previous study (referred to as numerical simulation 1, NS1, serving as the control simulation) and the new VEI (resulting in NS2, NS3, and NS4, representing the sensitivity simulations). The results indicated that the sensitivity simulations conducted using the BRAMS 5.2 model effectively captured PM₁₀ and PM_{2.5} concentrations, aligning closely with the measured data. The study provides an in-depth understanding of air quality dynamics that can assist in implementing strong policies and environmental management efforts to mitigate air pollution.

1. Introduction

With rapidly increasing urbanization in the Metropolitan Area of Lima – Callao (MALC), the vehicle population (VeP) in this region has increased from 1.5 million in 2013 to 2.1 million in 2021 (INEI, 2023). Vehicular emissions are the primary sources of ambient atmospheric aerosols (AAs) in urban areas (Gajbhiye et al., 2023; Hartikainen et al., 2023) that consequently exhibit negative consequences on economy and human health. Additionally, AAs can affect the local, regional, and global climate and weather (Bran et al.,

Abbreviations: VeP, vehicle population; AA, atmospheric aerosol; PM, particulate matter; CTM, chemical transport model; MALC, Metropolitan Area of Lima – Callao; BRAMS, Brazilian Regional Atmospheric Modeling System; VEI, vehicle emission inventory; AQMS, air quality monitoring stations; SB, San Borja; SJL, San Juan de Lurigancho; PP, Puente Piedra; MB, mean bias; ME, mean error; RMSE, root mean square error; NMB, normalized mean bias; NME, normalized mean error; FrB, fractional bias; FrE, fractional error; WRF-Chem, Weather Research and Forecasting model coupled to Chemistry; PRE-CHEM-SRC, preprocessor of trace gas and aerosol emission fields for regional and global atmospheric chemistry models; CO, carbon monoxide; NOx, nitrogen oxides; SO₂, sulfur dioxide; HC, Hydrocarbons; VOC, volatile organic compound..

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2022). Inhalable AA particulate matter with an aerodynamic diameter of less than or equal to $10\ \mu\text{m}$ is known as PM_{10} , while fine AA with an aerodynamic diameter of less than or equal to $2.5\ \mu\text{m}$ is known as $\text{PM}_{2.5}$. MALC is one of the most atmospheric polluted megacities in the Latin American and Caribbean Region (IQAir, 2022). Therefore, to ensure accurate atmospheric pollutant forecasting, using 3-dimensional Eulerian chemical transport models (CTMs), such as Brazilian Regional Atmospheric Modeling System (BRAMS) 5.2, is important. PM_{10} and $\text{PM}_{2.5}$ concentrations simulated with numerical models are lower using the vehicle emissions inventory (VEI) available in Lima in the entire Metropolitan Lima area and does not represent the spatiotemporal variations in vehicle emissions considered previously (Deuman and Walsh, 2005). This underpredicted AAs can be explained by the fact that VEIs have not been accurately estimated in MALC in previous studies. To solve this problem, we simulated PM_{10} and $\text{PM}_{2.5}$ concentrations using the BRAMS 5.2 system in Peru, which is the first study of its kind. The concentration simulations were input into the new VEI developed in this study in the preprocessor of trace gas and aerosol emission fields for regional and global atmospheric chemistry models (PRE-CHEM-SRC) system.

Predicting AAs efficiently is crucial to develop ambient particulate matter (PM) reduction regulations (Kong et al., 2021). The PM_{10} results simulated using the Weather Research and Forecasting model coupled to Chemistry (WRF-Chem) CTM in Lima, Peru, were lower than the measured PM_{10} concentrations (Sánchez-Ccoyllo et al., 2018). Incorrect emission inventory and representation of secondary AAs in the CTM (Bran et al., 2022), incorrect meteorological variables (Kitagawa et al., 2022), and incorrect chemical reactions and physical parameterizations in the CTM can generate different AA concentrations and their forecasts (Sha et al., 2022). To improve $\text{PM}_{2.5}$ forecasting in China, Kong et al. (2021) assimilated measured $\text{PM}_{2.5}$ data using the WRF-Chem model.

In addition, in China, Zhang et al. (2023) improved the prediction of safe $\text{PM}_{2.5}$ by coupling anthropogenic emissions with soil emissions from agricultural activity in the WRF-Chem model. Good air quality is beneficial for better public health, and thus, good air quality in MALC would attract further economic investments and tourists (Sharma et al., 2020). However, Xu and Qin (2023) found that implementing vehicle control policies contributed marginally to reducing AA emissions in Hanai Province, China.

Several research groups worldwide have performed air pollution forecasts using CTMs (Mogno et al., 2023; Sengupta et al., 2022) and statistical models (SMs) (Chauhan et al., 2021; Eren et al., 2023; Huang et al., 2021). However, SMs present the problem of incorrectly simulating air quality for extreme episodes of emissions and weather phenomena. Additionally, SMs do not accurately represent the relationship between primary and secondary pollutants (Baklanov and Zhang, 2020). Therefore, CTMs are more commonly used to understand the chemical-physical processes in air quality forecasting. Several studies have been conducted in Latin America and the Caribbean region on air quality using CTMs, particularly, in Colombia (Ballesteros-González et al., 2022; Lopez-Restrepo et al., 2020), Argentina (López-Noreña et al., 2022), and Brazil (Alonso et al., 2010; Freitas et al., 2017; Longo et al., 2013). However, numerical air pollution modeling using CTMs (Sánchez-Ccoyllo et al., 2022) has rarely been studied in Peru. Few studies have measured emission factors (EFs) of air pollution in South American megacities. For example, in São Paulo, the EF of AA was measured using road tunnel observations (Pereira et al., 2023; Sánchez-Ccoyllo et al., 2009). EF of urban pollutants is also measured using remote sensing system (Ghaffarpasand et al., 2023).

In this study, PM_{10} and $\text{PM}_{2.5}$ simulations were conducted using a numerical model coupled with air quality, carbon cycle, and meteorological conditions, called dynamic-downscaling BRAMS 5.2 system. The objective was to improve the simulations of PM_{10} and

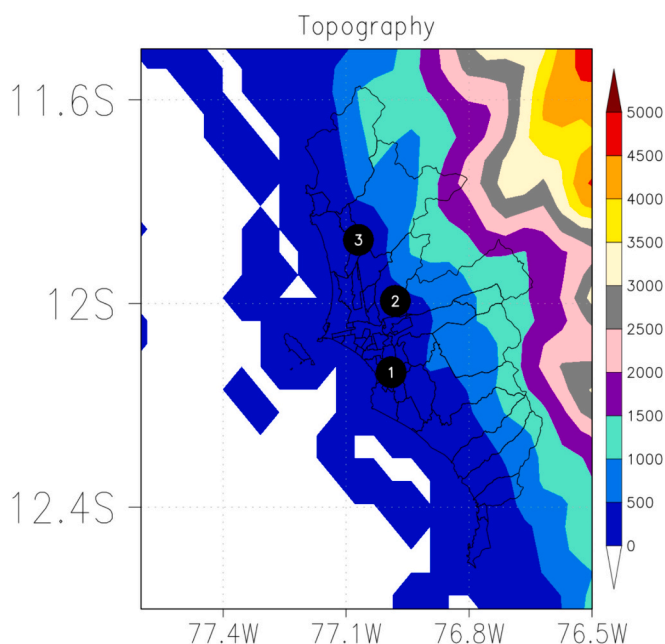


Fig. 1. Geographical area of PM_{10} and $\text{PM}_{2.5}$ simulations in Lima.

1: San Borja, 2: San Juan de Lurigancho, and 3: Puente Piedra

PM_{2.5} concentrations of MALC by performing four numerical simulations using the BRAMS model from February 1 to 6, 2018, using four types of VEIs developed in the PRE-CHEM-SRC system.

2. Materials and methods

2.1. Study area

MALC has three ambient air quality monitoring stations (AQMS) operated by SENAMHI (*Servicio Nacional de Meteorología e Hidrología del Perú; National Service of Meteorology and Hydrology of Peru*) to measure PM₁₀ and PM_{2.5} concentrations on an hourly basis. We selected only the available measured data of 75% (18 h of available data) of PM₁₀ and PM_{2.5} concentrations for each day; that is, the missing PM₁₀ and PM_{2.5} data can be up to 25% for each day (6 h) from February 1 to February 6, 2018. We chose the year 2018 due to its alignment with the most recent vehicle emissions report for the MALC. This report formed the cornerstone for us to develop a spatialized inventory. The primary objective of this study was to assess the effectiveness of the BRAMS modeling system in simulating Particulate Matter (PM) in the MALC. This involved presenting an updated and spatialized inventory of vehicle emissions (VEI), which underwent testing across various configurations. Notably, the findings indicated a substantial model sensitivity to the diurnal cycle of emissions. We opted to delve deeper into these uncertainties in the conclusion. The study focused on the dispersion of emission, justifying the analysis of a brief timeframe for this specific purpose. Future studies should evaluate the seasonal variability of PM₁₀ and PM_{2.5} with an improved modeling system (VEI and mathematical model). The AQMS were located in San Borja (SB), San Juan de Lurigancho (SJL), and Puente Piedra (PP) (Fig. 1). The SB AQMS (12° 6' 31.06" S; 77° 0' 27.96" W) is located 136 m asl, and is predominantly characterized by the presence of trees. The SJL AQMS (11° 58' 53.7" S, 76° 59' 57.6" W) is located 239 m asl, and is characterized predominantly by vehicular emissions. The PP AQMS (11° 51' 47.71" S, 77° 4' 26.88" W) is located 180 m asl, and is also characterized predominantly by vehicular emissions. In 2018, the average hourly PM_{2.5} and PM₁₀ concentrations reached 157.3 µg/m³ and 259.1 µg/m³, respectively, in the SJL AQMS. Moreover, on several days in 2018, the daily values of 15 µg/m³ for PM_{2.5} and 45 µg/m³ for PM₁₀ recommended by the World Health Organization were exceeded in Lima. In 2022, the air quality at SJL was the most polluted in the Latin America and Caribbean region, with the PM_{2.5} concentration being 41.2 µg/m³ (IQAir, 2022), which was eight times higher than the annual limit of 5 µg/m³ recommended by the World Health Organization under the new 2021 Annual Air Quality Guidelines.

On an urban scale, PM concentration exhibits high sensitivity to its sources. Among these sources, vehicular activity contributes significantly to both PM_{2.5} and PM₁₀. The processes of dispersion and deposition of these pollutants are influenced by variations in temperature and seasons. Our intention is to assess and analyze these processes more comprehensively in future studies. The mass concentration of PM₁₀ was measured using the Tapered Element Oscillating Microbalance (TEOM) 1405 Ambient Particulate Monitor. This gravimetric instrument operates by drawing ambient air through a filter at a consistent flow rate while continuously measuring the weight of the filter. It calculates near real-time mass concentrations of particulate matter. Furthermore, the instrument computes average PM₁₀ mass concentrations over different time intervals, including 1-h, 8-h, 12-h, and 24-h periods. The Model 5014i Continuous Ambient Particulate Monitor was employed to measure the mass concentration of PM_{2.5}. This instrument uses the radiometric principle of beta attenuation across a specified area on a fibrous filter tape to continuously ascertain the mass of ambient particles deposited. Through simultaneous refined mass measurements of sampled particulate on the filter tape and volume measurement of the sample, it enables a continuous concentration measurement of PM_{2.5} ambient particulate concentration.

2.2. Local emission factor and vehicle emissions inventory available in MALC

EF is the amount of emissions released into the atmosphere by human activities. In urban areas, EF is mainly associated with vehicular activities (Kumar et al., 2023; Song et al., 2021). According to Deuman and Walsh (2005), the EFs for each pollutant in MALC were Carbon Monoxide (CO) (14.6743 g km⁻¹), Nitrogen Oxides (NO_x) (5.58415 g km⁻¹), PM₁₀ (1.26019 g km⁻¹), Sulfur Dioxide (SO₂) (1.2645 g km⁻¹), and Hydrocarbons (HC) (4.67278 g km⁻¹), considering five types of vehicles, namely, light vehicles, taxis, buses, trucks, and motorcycles.

The VEI for Lima conducted by Deuman and Walsh (2005) includes CO (137,991 ton/yr), SO₂ (7022 ton/yr), NO_x (33,116 ton/yr), Volatile Organic Compound (VOC) (18,225 ton/yr), PM₁₀ (3271 ton/yr), and PM_{2.5} (2993 ton/yr). These VEI values were used in the first numerical simulation (NS) experiment (NS1) using the BRAMS 5.2 model.

2.3. Methodology for calculating the vehicle emissions in MALC

To improve air quality forecasting, using an inventory of reliable vehicle emissions is important. The emissions in MALC were calculated according to the method proposed by Zhu et al. (2023), as follows:

$$Et = VeP \times EF \times VKT \times 365 \times 10^{-6} \quad (1)$$

where

Et is the total vehicular emissions of air pollution in MALC (tons/yr), VeP is the annual vehicle population, EF is the vehicle emission factor of pollutants, measured in g km⁻¹, and VKT is the level of vehicle activity expressed in kilometers traveled.

In this study, the annual VeP automotive fleet in nationwide circulation in Peru was obtained from the website of *Instituto Nacional*

de Estadística e Informática (National Institute of Statistics and Informatics) (INEI, 2023). The EF values are described in Section 2.2. VKT is the average kilometers traveled per day for each vehicle category (Zhu et al., 2023). The VKT values in MALC were obtained from Lents et al. (2004).

2.4. PRE-CHEM-SRC setup in Lima

PRE-CHEM-SRC primarily consists of the Fortran 90 code, supplemented by a few C-code tools, which are responsible for providing $5 \text{ km} \times 5 \text{ km}$ horizontal gridded emissions data for atmospheric pollution in MALC. Subsequently, the emissions data were input for the air quality BRAMS 5.2 model. More details regarding PRE-CHEM-SRC are described in Freitas et al. (2011) and Longo et al. (2013). In this study, we employed four types of roads—primary, secondary, tertiary, and residential obtained from the Open Street Map dataset (Biljecki et al., 2023). Subsequently, the PRE-CHEM-SRC system was applied to estimate the CO, SO₂, NO_x, VOC, PM₁₀, and PM_{2.5} anthropogenic emissions with spatial and temporal resolutions of $5 \text{ km} \times 5 \text{ km}$ and 1 d, respectively (Fig. 2). The anthropogenic emissions inventory in MALC was developed in the PRE-CHEM-SRC pre-processor by combining the local emission inventory and the global inventories RETRO (REanalysis of the TROpospheric chemical composition over the past 40 years) according to previous studies (Alonso et al., 2010; Longo et al., 2013). The RETRO inventory provides worldwide emission information with a horizontal resolution of a $0.5^\circ \times 0.5^\circ$ grid for 26 species, as described in Bela et al. (2015).

2.5. BRAMS 5.2 model description and setup in MALC

The BRAMS 5.2 system is a numerical model integrated with atmospheric chemistry, the carbon cycle, and atmospheric circulation data (Freitas et al., 2017). In this study, the model employed a one-way nested grid configuration comprising a coarser grid with a horizontal resolution of 10 km and a finer nested grid with a resolution of 5 km, covering the MALC area. Simulations started on January 30, 2018 at 00 UTC. The model was run for 2 d for the spin-up. For the initial and boundary conditions of the meteorological variables of the first domain, we used the Final (FNL) global tropospheric analyses data from National Centers for Environmental Prediction (NCEP) with a horizontal resolution of $0.25^\circ \times 0.25^\circ$ and a temporal resolution of 6 h. Simulations in the second domain were performed using a grid with 30 longitude and latitude grid points. The model configurations are listed in Table 1. A more comprehensive description of the BRAMS 5.2 system can be found in previous studies (Freitas et al., 2017; Melo et al., 2019; Moreira et al., 2017).

The BRAMS model serves as the Brazilian adaptation of the RAMS (Regional Atmospheric Modeling System) model. Its online module for pollutants is Eurlen, predicting the mixing ratio of trace gases and aerosols by solving the conservation equation. This equation encompasses advection processes; turbulent mixing in the planetary boundary layer (PBL); dry deposition; elevation of plumes from fire outbreaks; and vertical transport associated with shallow (non-precipitating) and deep convection. These factors introduce uncertainties in pollutant modeling. Moreover, the model incorporates chemical reactivity and interactions of aerosols with solar and terrestrial radiation.

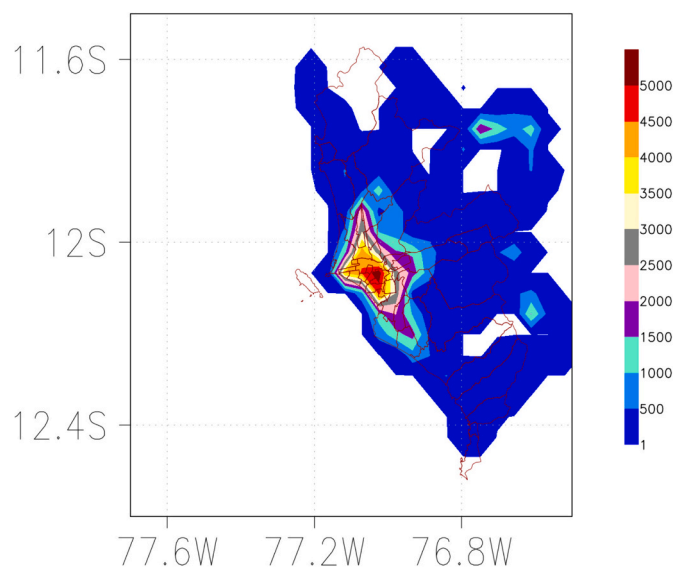


Fig. 2. Spatial distributions of CO emissions generated by the PRE-CHEM-SRC system (meters asl) in $\text{mg m}^{-2} \text{d}^{-1}$ on February 6, 2018. A colorful vertical legend visually represents the various CO emission values in $\text{mg m}^{-2} \text{d}^{-1}$.

Table 1
Selected configuration in the BRAMS 5.2 system for MALC.

Second domain simulation area	Metropolitan area of Lima-Callao
Simulation setup period	February 1–6, 2018
Model spin-up time	January 30–31, 2018
Domain center	–12.00° S, –77.00° W
Horizontal resolution	5 km × 5 km
Points in X and Y	30 and 30
Points in soil levels	7
Vertical resolution	33 vertical levels
Time step	8 s
Nudging in domain	Lateral is 3600 s (5 points), Top is 10,800 s (above of 15,000 m)
Aerosol processes	Simple monodisperse aerosol model (Longo et al., 2013)
Chemical processes	CCATT environmental model (Longo et al., 2013)
Radiation	RRTMG (Iacono et al., 2008)
Cumulus parametrization	Grell-Freitas scheme (Grell and Freitas, 2014)
Turbulence	Mellor–Yamada scheme (Mellor and Yamada, 1982)
Surface process	JULES (Moreira et al., 2013)

2.6. BRAMS 5.2 model evaluation

Hourly air pollution measurement data from the three AQMS (SB, SJL, and PP) were used to evaluate the model performance. Seven statistical metrics, namely, mean bias (MB), mean error (ME), root-mean-square error (RMSE), normalized mean bias (NMB), normalized mean error (NME), fractional bias (FrB), and fractional error (FrE), were used to evaluate the model performance. The calculation formulas for these metrics are given below.

$$MB = \frac{1}{N} \sum_{k=1}^N (B_k - O_k) \quad (2)$$

$$ME = \frac{1}{N} \sum_{k=1}^N |B_k - O_k| \quad (3)$$

$$RMSE = \sqrt{\sum_{k=1}^N (B_k - O_k)^2 \frac{1}{N}} \quad (4)$$

$$MNB = \frac{\sum_{k=1}^N (B_k - O_k)}{\sum_{k=1}^N O_k} \times 100 \quad (5)$$

$$NME = \frac{\sum_{k=1}^N |B_k - O_k|}{\sum_{k=1}^N O_k} \times 100 \quad (6)$$

$$FrB = \frac{2}{N} \sum_{k=1}^N \frac{(B_k - O_k)}{(B_k + O_k)} \times 100 \quad (7)$$

$$FrE = \frac{2}{N} \sum_{k=1}^N \frac{|B_k - O_k|}{(B_k + O_k)} \times 100 \quad (8)$$

where

B_k and O_k are the predicted and observed PM_{10} and $PM_{2.5}$ concentrations, respectively, and N is the hourly PM_{10} and $PM_{2.5}$ concentration observed-predicted pair.

3. Results and discussion

3.1. New inventory of vehicular emissions in MALC

The new VEI resulted from a modification of the Fortran program called retroemissions.f90. Specifically, we incorporated the total

annual VEI data (estimated using Eq. 1) for the entire MALC area and further allocated them into primary, secondary, tertiary, and resident categories based on vehicle distribution across these four road types, according to the methodology outlined in Lents et al. (2004). This newly created VEI was a key component of the PRE-CHEM-SRC tool described in Section 2.4, and was used for NS2 (Table 2).

To conduct NS3, the original $PM_{2.5}$ VEI was calculated as 60% of the total PM_{10} emissions, and the emissions of other air pollutants of CO, SO_2 , NO_x , VOC, and PM_{10} were maintained as the ones mentioned in Table 2, and are shown in Table 3. The NS3 was made using the ratio of emission $PM_{2.5}/PM_{10}$ were obtained from Lents et al. (2004).

To conduct NS4, the new $PM_{2.5}$ VEI was calculated as 38% of the total PM_{10} emissions, and the inventory values for vehicle emissions of CO, SO_2 , NO_x , VOC, and PM_{10} were the same as those described in Table 2 and are presented in Table 4. It is noteworthy that the NS4 study was conducted solely with the aim of enhancing simulated $PM_{2.5}$ concentrations. In pursuit of this goal, the correlations between $PM_{2.5}$ and PM_{10} concentrations recorded in the atmosphere of the metropolitan area of Lima and Callao from 2018 to 2019 were computed. The resulting value for this correlation, termed as the $PM_{2.5}/PM_{10}$ ratio, was determined to be 0.36.

3.2. PM simulated with the BRAMS model in MALC

Fig. 3 shows a comparison of hourly concentrations of measured $PM_{2.5}$ ($PM_{2.5}$.Obs) and the results of four types of numerical simulations of $PM_{2.5}$ using the BRAMS model. NS1 is the control simulation of $PM_{2.5}$ concentrations ($PM_{2.5}$.NS1 in red) and was performed with input to the BRAMS 5.2 model from the VEI conducted by Deuman and Walsh (2005) in MALC. Very low simulated $PM_{2.5}$ concentrations were observed with respect to $PM_{2.5}$ measurements in the NS1 results. This occurred because the VEI conducted by Deuman and Walsh (2005) is outdated and apparently has very low values in MALC. According to Shin et al. (2021), the uncertainty in VEI is because of the inadequate calculation of anthropogenic sources and incorrect use of EF. NS2 as a sensitivity simulation of $PM_{2.5}$ concentrations ($PM_{2.5}$.NS2 in blue) was the new VEI calculated in this study (as explained in Section 3.1) and was used as an input to the BRAMS model. NS3 as a sensitivity simulation ($PM_{2.5}$.NS3 in pink) showed that the simulated $PM_{2.5}$ concentrations were lower than those by NS2, but the $PM_{2.5}$ values were higher than the $PM_{2.5}$ observations. Finally, NS4 as a sensitivity simulation of $PM_{2.5}$ concentrations ($PM_{2.5}$.NS4 in violet) showed values of the simulated hourly concentrations of $PM_{2.5}$ similar to the measured $PM_{2.5}$ concentrations. These results indicated improved simulation of $PM_{2.5}$ concentrations in MALC. The simulated $PM_{2.5}$ diurnal cycle in China as performed by Sha et al. (2022) using the WRF-Chem model version 3.9.1 was higher than the $PM_{2.5}$ concentration measured in the morning and evening by a factor of 2.5.

It's plausible that the instruments employed for $PM_{2.5}$ measurements in the real-world data, such as the Model 5014i Continuous Ambient Particulate Monitor, possess inherent limitations that hinder the accurate capture of the peak witnessed in the simulation (Fig. 3). These constraints may arise from various factors including instrument sensitivity, sampling methodologies like beta attenuation, or inherent measurement inaccuracies. Consequently, these limitations could contribute to the disparities observed between the simulated and measured $PM_{2.5}$ concentrations on day four (Fig. 3). Notably, it's worth mentioning that despite this discrepancy, both the measured data obtained using the TEOM 1405 Ambient Particulate Monitor and the simulated data for PM_{10} in Fig. 4 depict the presence of the peak on day four.

Fig. 4 shows the comparisons between the measured hourly concentrations of PM_{10} (PM_{10} .Obs) and the hourly PM_{10} concentration simulated using the BRAMS model. Four numerical simulations of PM_{10} concentrations were performed. NS1 of the PM_{10} concentrations (PM_{10} .NS1) was conducted using the input of the VEI available in MALC from a previous study into the BRAMS model. The BRAMS model results for this case indicated extremely low concentrations of PM_{10} (approximately $4 \mu g/m^3$) compared to their PM_{10} measurements (average for the study period = $77.1 \mu g/m^3$). The VEI conducted by Deuman and Walsh (2005) was extremely low in MALC, which is the official emission inventory. NS2, NS3, and NS4 were performed using the new VEI calculated in the present study as input to the BRAMS model. The simulation results of PM_{10} concentrations for NS4 were similar to those of the PM_{10} measurements. Thus, the simulations of PM_{10} concentrations were satisfactorily improved in MALC.

The emission values for CO, SO_2 , NO_x , VOC, and PM_{10} remain consistent across NS2, NS3, and NS4. However, there is a variation in the $PM_{2.5}$ emission values between these simulations. Consequently, the concentrations of PM_{10} are identical for NS2, NS3, and NS4. Therefore, Fig. 4 is not included for NS2 and NS3, but it exclusively appears for NS4.

The PM_{10} and $PM_{2.5}$ concentrations predicted using the BRAMS 5.2 system were compared with the observed PM_{10} and $PM_{2.5}$ concentrations, to evaluate the performance of the BRAMS 5.2 system (Table 5). The MB for $PM_{2.5}$ increased from $-17.8 \mu g/m^3$ for NS1 to $32.7 \mu g/m^3$ for NS2, and then decreased to $0.8 \mu g/m^3$ in NS4. Moreover, the ME and RMSE of $PM_{2.5}$ decreased from 17.8 and 18.8

Table 2

New inventory of vehicle emissions in Metropolitan Lima (ton/yr) calculated using the average emission factor for five types of vehicles used in NS2.

Pollutants	Total emission	Primary	Secondary	Tertiary	Resident
CO	767,223.0	515,497.1	96,900.3	96,900.3	57,925.3
SO_2	66,112.0	44,420.7	8349.9	8349.9	4991.5
NO_x	291,959.0	196,167.3	36,874.4	36,874.4	22,042.9
VOC	244,309.0	164,151.	30,856.2	30,856.2	18,445.3
PM_{10}	65,887.0	44,269.5	8321.5	8321.5	4974.5
$PM_{2.5}$ ¹	59,298.0	39,842.3	7489.3	7489.3	4477.0

¹ $PM_{2.5}$ is 90% of PM_{10} emissions.

Table 3

New inventory of vehicle emissions (ton/yr) is the same as that described in Table 2, but only PM_{2.5} emissions were calculated as 60% of PM₁₀ emissions.

Pollutants	Total Emission	Primary	Secondary	Tertiary	Resident
CO	767,223.0	515,497.1	96,900.3	96,900.3	57,925.3
SO ₂	66,112.0	44,420.7	8349.9	8349.9	4991.5
NO _x	291,959.0	196,167.3	36,874.4	36,874.4	22,042.9
VOC	244,309.0	164,151.	30,856.2	30,856.2	18,445.3
PM ₁₀	65,887.0	44,269.5	8321.5	8321.5	4974.5
PM _{2.5} ¹	39,532.2	26,561.7	4992.9	4992.9	2984.7

¹ PM_{2.5} is 60% of PM₁₀ emissions.

Table 4

New vehicle emissions inventory (ton/yr) is the same as that described in Table 2, but PM_{2.5} emissions were calculated as 38% of PM₁₀ emissions.

Pollutants	Total Emission	Primary	Secondary	Tertiary	Resident
CO	767,223.0	515,497.1	96,900.3	96,900.3	57,925.3
SO ₂	66,112.0	44,420.7	8349.9	8349.9	4991.5
NO _x	291,959.0	196,167.3	36,874.4	36,874.4	22,042.9
VOC	244,309.0	164,151.	30,856.2	30,856.2	18,445.3
PM ₁₀	65,887.0	44,269.5	8321.5	8321.5	4974.5
PM _{2.5} ¹	23,719.0	15,936.8	2995.7	2995.7	1790.8

¹ PM_{2.5} is 38% of PM₁₀ emissions.

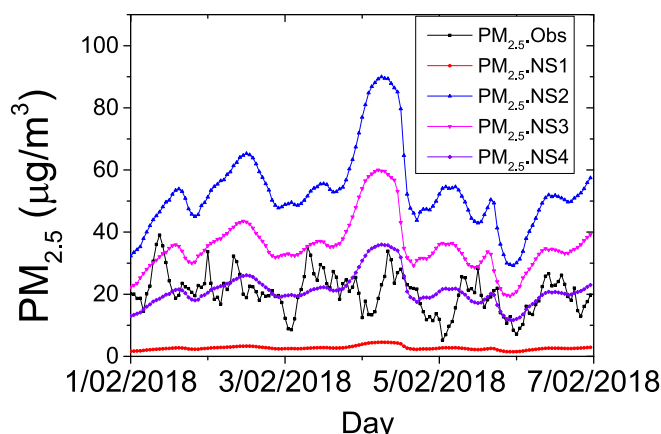


Fig. 3. Time-series of the measured hourly PM_{2.5} concentrations (PM_{2.5}.Obs) comparing the outputs of the BRAMS model simulations that used four types of emission inventories in MALC as input: i) Official emission inventory (PM_{2.5}.NS1), ii) PM_{2.5} emission is 90% of PM₁₀ (PM_{2.5}.NS2), iii) PM_{2.5} emission is 60% of PM₁₀ (PM_{2.5}.NS3), and iv) PM_{2.5} emission is 38% of PM₁₀ (PM_{2.5}.NS4).

µg/m³ obtained in NS1 to 5.3 and 7.2 µg/m³ obtained in NS4, respectively. For PM_{2.5}, the NMB increased from −86.9% for NS1 to 159.9% for NS2, and then decreased to 4.0% obtained in NS4, which met the reliable PM_{2.5} simulation target suggested by Emery et al. (2017), who suggested an NMB value of < ±10% for PM_{2.5}. Further, the NME for PM_{2.5} increased from 86.9% for NS1 to 159.9% for NS2, and then decreased to 26.1% obtained in NS4, which also met the reliable simulation target of <35%, as suggested by Emery et al. (2017). The FrB for PM_{2.5} increased from −150.3% in NS1 to 88.8% in NS2 and then decreased to 6.2% obtained in NS4. Moreover, the FrE value of PM_{2.5} decreased from −150.3% obtained in NS1 to 25.9% obtained in NS4.

Results shown in bold met the goal for PM_{2.5} recommended by Boylan and Russell (2006) and Emery et al. (2017).

It's important to note that NS2 and NS3 serve as intermediate steps toward achieving the desired NS4 (Table 5), characterized by outstanding statistical metrics for PM_{2.5}. Specifically, metrics such as Normalized Mean Bias (%), Normalized Mean Error (%), Fractional Bias (%), and Fractional Error (%) have been meticulously optimized to meet the criteria recommended by Boylan and Russell (2006) and Emery et al. (2017). For PM_{2.5}, these metrics stand at 4.0, 26.1, 6.2, and 25.9, respectively, thus satisfying the predefined benchmarks.

The MB, ME, and RMSE values of PM₁₀ decreased from 70.5, 70.5, and 72.8 µg/m³ obtained in NS1 to −14.4, 15.9, and 20.0 µg/m³ obtained in NS4, respectively. The NMB and NME values of PM₁₀ decreased from −96.0 and 96.0 µg/m³ obtained in NS1 to −19.6 and 21.7 µg/m³ obtained in NS4, respectively. Further, the FrB and FrE values decreased from −184.3 and 184.3 µg/m³ obtained in NS1 to −21.0 and 23.2 µg/m³ obtained in NS4, respectively.

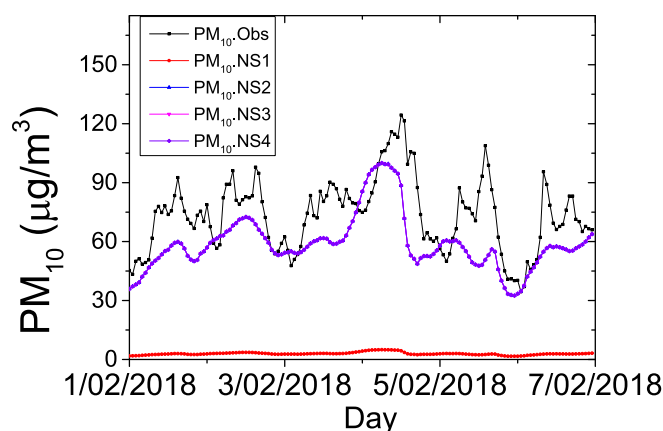


Fig. 4. Time-series of the measured hourly PM_{10} concentrations (PM_{10} -Obs) compared to the outputs of the BRAMS model simulations used for the input of four types of emission inventories in MALC: Official emission inventory (PM_{10} -NS1) and the new VEI calculated in this study. NS1 is numerical simulation 1; NS2 is numerical simulation 2; NS3 is numerical simulation 3; and, NS4 is numerical simulation 4.

Table 5

Comparison of atmospheric aerosol observations in Lima City.

Statistical measure	Numerical simulation 1	Numerical simulation 2	Numerical simulation 3	Numerical simulation 4
$PM_{2.5}$				
Mean bias ($\mu\text{g}/\text{m}^3$)	-17.8	32.7	15.1	0.8
Mean error ($\mu\text{g}/\text{m}^3$)	17.8	32.7	15.3	5.3
RMSE ($\mu\text{g}/\text{m}^3$)	18.8	35.2	17.2	7.2
Normalized mean bias (%)	-86.9	159.9	73.9	4.0
Normalized mean error (%)	86.9	159.9	74.9	26.1
Fractional bias (%)	-150.3	88.8	54.8	6.2
Fractional error (%)	150.3	88.8	55.4	25.9
PM_{10}				
Mean bias ($\mu\text{g}/\text{m}^3$)	-70.5	-14.4	-14.4	-14.4
Mean error ($\mu\text{g}/\text{m}^3$)	70.5	15.9	15.9	15.9
RMSE ($\mu\text{g}/\text{m}^3$)	72.8	20.0	20.0	20.0
Normalized mean bias (%)	-96.0	-19.6	-19.6	-19.6
Normalized mean error (%)	96.0	21.7	21.7	21.7
Fractional bias (%)	-184.3	-21.0	-21.0	-21.0
Fractional error (%)	184.3	23.2	23.2	23.2

Fig. 5 presents a comparison between the measured wind rose plot (Fig. 5a) and the predicted wind rose plot using the BRAMS 5.2 model (Fig. 5b) at the SJL station located in the eastern zone of MALC. The measured and simulated wind directions were obtained from the south-southwest (SSW) direction with wind speeds varying from 2 to 4 m/s for both the measured and simulated wind directions. The measured and simulated SSW wind directions were observed for 60% and 55% of the time during the study period. Moreover, the measured and simulated wind directions had a predominantly south wind direction for 25% and 40% of the time with a south direction, respectively. The measured and simulated wind speeds varied from 4 to 6 m/s. The results of the wind direction and wind speed simulations using the BRAMS 5.2 model reproduced the measured wind speed and direction.

The highest values of $PM_{2.5}$ concentrations predicted using the BRAMS model were observed in the north and east of MALC (Fig. 6). This was because $PM_{2.5}$ was transported by wind from the center and south of MALC. A predominant wind direction simulated at MALC was from the south (S) and SSW (Fig. 6). The average observed and simulated $PM_{2.5}$ concentrations were 20.3 and 22.5 $\mu\text{g}/\text{m}^3$, respectively, which were similar in MALC.

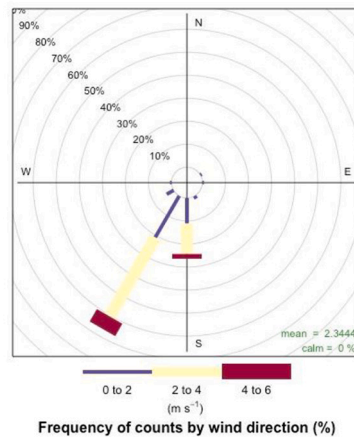
Elevated simulated PM_{10} concentrations were also observed to the north and east of MALC. The predicted wind direction was from the south and SSW (Fig. 7). These predominant wind directions could transport PM_{10} from areas in central Lima to northern and eastern Lima. Conversely, in the entire MALC area, the average predicted and average PM_{10} concentrations were 77.5 and 62.4 $\mu\text{g}/\text{m}^3$, respectively, during the simulation period. The simulated PM_{10} concentrations were similar to the measured PM_{10} concentrations.

The vertical color scale shows the $PM_{2.5}$ concentrations values in $\mu\text{g}/\text{m}^3$.

4. Conclusions

This study aimed to improve simulations of PM_{10} and $PM_{2.5}$ concentrations using a numerical air quality model, BRAMS 5.2. Based on the new VEI calculated in this study, emissions were spatially distributed considering four street types (primary, secondary, tertiary, and resident) and the temporal distribution of daily emissions. A Fortran program called retroemissions.f90 in the PRE-CHEM-SRC

a) Observed



b) Simulated

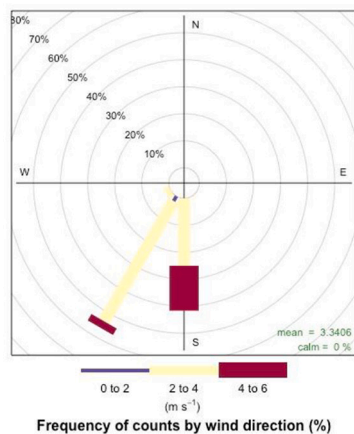


Fig. 5. Wind rose plot of wind direction and speed based on the in-situ observations (a) and BRAMS 5.2 model (b). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

software was modified. Simulations were conducted using the BRAMS 5.2 system with the input of these updated VEIs. The results indicated significant improvements in the accuracy of PM_{10} and $PM_{2.5}$ simulations. These improvements were quantitatively confirmed by comparing the model results with the measured PM_{10} and $PM_{2.5}$ concentrations. Specifically, the ME for $PM_{2.5}$ decreased from 17.8 to 5.3 $\mu\text{g}/\text{m}^3$, RMSE improved from 18.8 to 7.2 $\mu\text{g}/\text{m}^3$, and FrE reduced from 150.3 to 25.9 $\mu\text{g}/\text{m}^3$, when the results were compared to the simulations acquired in a previous study using vehicle emissions data. Similarly, for PM_{10} , a noticeable reduction in the ME from 70.5 to 15.9 $\mu\text{g}/\text{m}^3$ was recorded, while the RMSE improved from 72.8 to 20.0 $\mu\text{g}/\text{m}^3$, and FrE decreased from 184.3 to 23.2 $\mu\text{g}/\text{m}^3$, on comparison with the vehicle emission data of previous studies. In conclusion, these findings underscore the substantial enhancements achieved in PM_{10} and $PM_{2.5}$ simulations, aligning them more closely with the observed concentration measurements in MALC, owing to the implementation of an updated VEI and the modified PRE-CHEM-SRC system.

The findings of this study highlight the high sensitivity of the model to emissions, emphasizing the need for a more detailed examination of the diurnal cycle of the vehicle fleet in MALC. This could involve vehicle counting or using coupled models incorporating radar networks. Additionally, we recommend to conduct further testing with the chemical module to incorporate secondary aerosol formation in BRAMS. Lastly, the article strongly suggests future research involving numerical simulations of PM_{10} and $PM_{2.5}$ using alternative air quality numerical models, such as WRF-Chem.

Future studies should also incorporate the newly developed VEIs established in this study. The present study provides valuable insights and validation of the effectiveness of emissions inventory improvements in enhancing the accuracy of PM_{10} and $PM_{2.5}$ simulations. Expanding the scope of modeling by employing different numerical models can contribute to a more comprehensive understanding of air quality dynamics and support more robust policymaking and environmental management efforts.

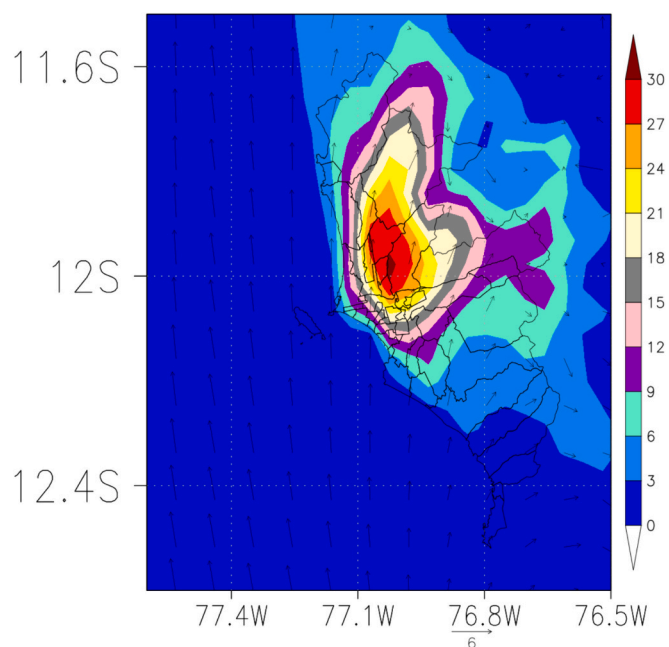


Fig. 6. Spatial distribution of simulated $PM_{2.5}$ concentrations ($\mu\text{g}/\text{m}^3$) based on NS4 in MALC. The vertical color scale shows the $PM_{2.5}$ concentrations values in $\mu\text{g}/\text{m}^3$.

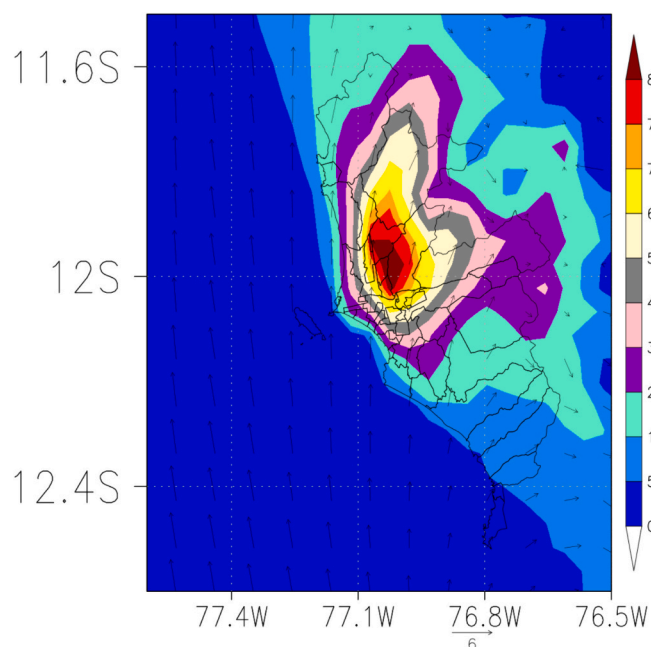


Fig. 7. Spatial distribution of PM_{10} concentrations predicted in MALC.

Consent for publication

All authors consent to publish this article in Urban Climate.

CRediT authorship contribution statement

Odón R. Sánchez-Ccoyllo: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal

analysis, Data curation, Conceptualization. **Marcelo Alonso:** Writing – review & editing, Visualization, Software, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could influence the work reported in this manuscript.

Data availability

Data will be made available on request.

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References

- Alonso, M.F., Longo, K.M., Freitas, S.R., Fonseca, M.R., Marécal, V., Pirre, M., Klenner, L.G., 2010. An urban emissions inventory for South America and its application in numerical modeling of atmospheric chemical composition at local and regional scales. *Atmos. Environ.* 44 (39), 5072–5083. <https://doi.org/10.1016/j.atmosenv.2010.09.013>.
- Baklanov, A., Zhang, Y., 2020. Advances in air quality modeling and forecasting. *Glob. Transit.* 2, 261–270. <https://doi.org/10.1016/j.glt.2020.11.001>.
- Ballesteros-González, K., Espitia-Cano, S.O., Rincón-Caro, M.A., Rincón-Riveros, J.M., Perez-Peña, M.P., Sullivan, A., Morales Betancourt, R., 2022. Understanding organic aerosols in Bogotá, Colombia: in-situ observations and regional-scale modeling. *Atmos. Environ.* 284, 119161 <https://doi.org/10.1016/j.atmosenv.2022.119161>.
- Bela, M.M., Longo, K.M., Freitas, S.R., Moreira, D.S., Beck, V., Wofsy, S.C., Gerbig, C., Wiedemann, K., Andreae, M.O., Artaxo, P., 2015. Ozone production and transport over the Amazon Basin during the dry-to-wet and wet-to-dry transition seasons. *Atmos. Chem. Phys.* 15 (2), 757–782. <https://doi.org/10.5194/acp-15-757-2015>.
- Biljecki, F., Chow, Y.S., Lee, K., 2023. Quality of crowdsourced geospatial building information: a global assessment of OpenStreetMap attributes. *Build. Environ.* 237, 110295 <https://doi.org/10.1016/j.buildenv.2023.110295>.
- Boylan, J.W., Russell, A.G., 2006. PM and light extinction model performance metrics, goals, and criteria for three-dimensional air quality models. *Atmos. Environ.* 40, 4946–4959. <https://doi.org/10.1016/j.atmosenv.2005.09.087>.
- Bran, S.H., Macatangay, R., Surapipith, V., Chotamonsak, C., Chantara, S., Han, Z., Li, J., 2022. Surface PM_{2.5} mass concentrations during the dry season over northern Thailand: sensitivity to model aerosol chemical schemes and the effects on regional meteorology. *Atmos. Res.* 277, 106303 <https://doi.org/10.1016/j.atmosres.2022.106303>.
- Chauhan, R., Kaur, H., Alankar, B., 2021. Air quality forecast using convolutional neural network for sustainable development in urban environments. *Sustain. Cities Soc.* 75, 103239 <https://doi.org/10.1016/j.scs.2021.103239>.
- Deuman, I., Walsh, P., 2005. Environmental Baseline Study COSAC I. Deuman International/Walsh Perú Ingenieros y Científicos Consultores, Lima. <https://www.protransporte.gob.pe/pdf/biblioteca/2005/lineabaseambiental/capitulo10.pdf> (accessed 25 July 2022).
- Emery, C., Liu, Z., Russell, A.G., Odmann, M.T., Yarwood, G., Kumar, N., 2017. Recommendations on statistics and benchmarks to assess photochemical model performance. *J. Air Waste Manage. Assoc.* 67 (5), 582–598. <https://doi.org/10.1080/10962247.2016.1265027>.
- Eren, B., Aksangür, I., Erden, C., 2023. Predicting next hour fine particulate matter (PM_{2.5}) in the Istanbul Metropolitan City using deep learning algorithms with time windowing strategy. *Urban Clim.* 48, 101418 <https://doi.org/10.1016/j.uclim.2023.101418>.
- Freitas, S.R., Longo, K.M., Alonso, M.F., Pirre, M., Marecal, V., Grell, G., Stockler, R., Mello, R.F., Sánchez Gácita, M., 2011. PREP-CHEM-SRC - 1.0: a preprocessor of trace gas and aerosol emission fields for regional and global atmospheric chemistry models. *Geosci. Model Dev.* 4, 419–433. <https://doi.org/10.5194/gmd-4-419-2011>.
- Freitas, S.R., Panetta, J., Longo, K.M., Rodrigues, L.F., Moreira, D.S., Rosário, N.E., Silva Dias, P.L., Silva Dias, M.A.F., Souza, E.P., Freitas, E.D., Longo, M., Frasson, A., Fazenda, A.L., Santos E Silva, C.M., Pavani, C.A.B., Eiras, D., França, D.A., Massaru, D., Silva, F.B., Santos, F.C., Pereira, G., Camponogara, G., Ferrada, G.A., Velho, H.F.C., Menezes, I., Freire, J.L., Alonso, M.F., Gacita, M.S., Zarzur, M., Fonseca, R.M., Lima, R.S., Siqueira, R.A., Braz, R., Tomita, S., Oliveira, V., Martins, L.D., 2017. The Brazilian developments on the regional atmospheric modeling system (BRAMS 5.2): an integrated environmental model tuned for tropical areas. *Geosci. Model Dev.* 10, 189–222. <https://doi.org/10.5194/gmd-10-189-2017>.
- Gajbhiye, M.D., Lakshmanan, S., Aggarwal, R., Kumar, N., Bhattacharya, S., 2023. Evolution and mitigation of vehicular emissions due to India's Bharat stage emission standards – a case study from Delhi. *Environ. Develop.* 45, 100803 <https://doi.org/10.1016/j.envdev.2023.100803>.
- Ghaffarpasand, O., Ropkins, K., Beddows, D.C.S., Pope, F.D., 2023. Detecting high emitting vehicle subsets using emission remote sensing systems. *Sci. Total Environ.* 858, 159814 <https://doi.org/10.1016/j.scitotenv.2022.159814>.
- Grell, G.A., Freitas, S.R., 2014. A scale and aerosol aware stochastic convective parameterization for weather and air quality modeling. *Atmos. Chem. Phys.* 14 (10), 5233–5250. <https://doi.org/10.5194/acp-14-5233-2014>.
- Hartikainen, A.H., Ihalainen, M., Yli-Pirilä, P., Hao, L., Kortelainen, M., Pieber, S.M., Sippula, O., 2023. Photochemical transformation and secondary aerosol formation potential of Euro6 gasoline and diesel passenger car exhaust emissions. *J. Aerosol Sci.* 171, 106159 <https://doi.org/10.1016/j.jaerosci.2023.106159>.
- Huang, G., Li, X., Zhang, B., Ren, J., 2021. PM_{2.5} concentration forecasting at surface monitoring sites using GRU neural network based on empirical mode decomposition. *Sci. Total Environ.* 768, 144516 <https://doi.org/10.1016/j.scitotenv.2022.144516>.
- Iacono, M.J., Delamere, J.S., Mlawer, E.J., Shephard, M.W., Clough, S.A., Collins, W.D., 2008. Radiative forcing by long-lived greenhouse gases: calculations with the AER radiative transfer models. *J. Geophys. Res. Atmos.* 113 (13) <https://doi.org/10.1029/2008JD009944>.
- INEI, 2023. National Institute of Statistics and Informatics of Peru. <https://www.gob.pe/inei/> (accessed 21 June 2022).
- IQAir, 2022. 2022 World Air Quality Report. Region and City PM_{2.5} Ranking. <https://www.iqair.com/world-air-quality-report> (accessed 15 April 2023).
- Kitagawa, Y.K., Albuquerque, A.T.T., Kumar, P., Nascimento, E.G.S., Moreira, D.M., 2022. Coastal-urban meteorology: a sensitivity study using the WRF-urban model. *Urban Clim.* 44, 101185 <https://doi.org/10.1016/j.uclim.2022.101185>.
- Kong, Y., Sheng, L., Li, Y., Zhang, W., Zhou, Y., Wang, W., Zhao, Y., 2021. Improving PM_{2.5} forecast during haze episodes over China based on a coupled 4D-LETKF and WRF-Chem system. *Atmos. Res.* 249, 105366 <https://doi.org/10.1016/j.atmosres.2020.105366>.
- Kumar, P., Beig, G., Sahu, S.K., Yadav, R., Maji, S., Singh, V., Bamniya, B.R., 2023. Development of a high-resolution emissions inventory of carbonaceous particulate matters and their growth during 2011–2018 over India. *Atmos. Environ.* 303, 119750 <https://doi.org/10.1016/j.atmosenv.2023.119750>.
- Lents, J., Davis, N., Nikkila, N., Osses, M., 2004. Lima Vehicle Activity Study. Final report. <http://www.gssr.net/>.

- Longo, K.M., Freitas, S.R., Pirre, M., Marécal, V., Rodrigues, L.F., Panetta, J., Alonso, M.F., Rosário, N.E., Moreira, D.S., Gácita, M.S., Arteta, J., Fonseca, R., Stockler, R., Katsurayama, D.M., Fazenda, A., Bela, M., 2013. The chemistry CATT-BRAMS model (CCATT-BRAMS 4.5): a regional atmospheric model system for integrated air quality and weather forecasting and research. *Geosci. Model Dev.* 6 (5), 1389–1405. <https://doi.org/10.5194/gmd-6-1389-2013>.
- López-Noreña, A.I., Berná, L., Tames, M.F., Millán, E.N., Puliafito, S.E., Fernandez, R.P., 2022. Influence of emission inventory resolution on the modeled spatio-temporal distribution of air pollutants in Buenos Aires, Argentina, using WRF-Chem. *Atmos. Environ.* 269, 118839 <https://doi.org/10.1016/j.atmosenv.2021.118839>.
- Lopez-Restrepo, S., Yarce, A., Pinel, N., Quintero, O.L., Segers, A., Heemink, A.W., 2020. Forecasting PM10 and PM2.5 in the Aburrá Valley (Medellín, Colombia) via EnKF based data assimilation. *Atmos. Environ.* 232, 117507 <https://doi.org/10.1016/j.atmosenv.2020.117507>.
- Mellor, G.L., Yamada, T., 1982. Development of a turbulence closure model for geophysical fluid problems. *Rev. Geophys.* 20 (4), 851–875. <https://doi.org/10.1029/RG020i004p00851>.
- Melo, A.M.Q., Dias-Junior, C.Q., Cohen, J.C.P., Sá, L.D.A., Cattaneo, J.H., Kuhn, P.A.F., 2019. Ozone transport and thermodynamics during the passage of squall line in Central Amazon. *Atmos. Environ.* 206, 132–143. <https://doi.org/10.1016/j.atmosenv.2019.02.018>.
- Mogno, C., Palmer, P.I., Marvin, M.R., Sharma, S., Chen, Y., Wild, O., 2023. Road transport impact on PM2.5 pollution over Delhi during the post-monsoon season. *Atmos. Environ.* X, 17, 100200 <https://doi.org/10.1016/j.aeaoa.2022.100200>.
- Moreira, D.S., Freitas, S.R., Bonatti, J.P., Mercado, L.M., Rosário, E., N.M., Longo, K.M., Miller, J.B., Gloor, M., Gatti, L.V., 2013. Coupling between the JULES land-surface scheme and the CCATT-BRAMS atmospheric chemistry model (JULES-CCATT-BRAMS1.0): applications to numerical weather forecasting and the CO2 budget in South America. *Geosci. Model Dev.* 6 (4), 1243–1259. <https://doi.org/10.5194/gmd-6-1243-2013>.
- Moreira, D.S., Longo, K.M., Freitas, S.R., Yamasoe, M.A., Mercado, L.M., Rosário, N.E., Gloor, E., Viana, R.S.M., Miller, J.B., Gatti, L.V., Wiedemann, K.T., Domingues, L.K.G., Correia, C.C.S., 2017. Modeling the radiative effects of biomass burning aerosols on carbon fluxes in the Amazon region. *Atmos. Chem. Phys.* 17 (23), 14785–14810. <https://doi.org/10.5194/acp-17-14785-2017>.
- Pereira, G.M., Nogueira, T., Kamigauti, L.Y., Santos, M.D., Nascimento, E.Q.M., Martins, J.V., Vicente, A., Artaxo, P., Alves, C., Vasconcellos, C.P., Andrade, F.M., 2023. Particulate matter fingerprints in biofuel impacted tunnels in South America's largest metropolitan area. *Sci. Total Environ.* 856, 159006 <https://doi.org/10.1016/j.scitotenv.2022.159006>.
- Sánchez-Ccoyllo, O.R., Ynoue, R.Y., Martins, L.D., Astolfo, R., Miranda, R.M., Freitas, E.D., Borges, A.S., Fornaro, A., Freitas, H., Moreira, A., Andrade, M.F., 2009. Vehicular particulate matter emissions in road tunnels in Sao Paulo, Brazil. *Environ. Monit. Assess.* 149 (1–4), 241–249. <https://doi.org/10.1007/s10661-008-0198-5>.
- Sánchez-Ccoyllo, O.R., Ordoñez-Aquino, C.G., Muñoz, Á.G., Llacza, A., Andrade, M.F., Liu, Y., Reátegui-Romero, W., Brasseur, G., 2018. Modeling study of the particulate matter in lima with the WRF-Chem model: case study of April 2016. *Int. J. Appl. Eng. Res.* 13, 10129–10141. <https://doi.org/10.37622/ijaer/13.11.2018.10129-10141>.
- Sánchez-Ccoyllo, O.R., Llacza, A., Ayma-Choque, E., Alonso, M., Castesana, P., de Andrade, M.F., 2022. Evaluating the impact of vehicular aerosol emissions on particulate matter (PM2.5) formation using modeling study. *Atmosphere* 13 (11), 1816. <https://doi.org/10.3390/atmos13111816>.
- Sengupta, A., Govardhan, G., Deb Nath, S., Yadav, P., Kulkarni, S.H., Parde, A.N., Lonkar, P., Dhangar, N., Gunwani, P., Wagh, S., Nivdange, S., Jena, C., Kumar, R., Ghude, S.D., 2022. Probing into the wintertime meteorology and particulate matter (PM2.5 and PM10) forecast over Delhi. *Atmos. Pollut. Res.* 13 (6), 101426 <https://doi.org/10.1016/j.apr.2022.101426>.
- Sha, T., Ma, X., Wang, J., Tian, R., Zhao, J., Cao, F., Zhang, Y.L., 2022. Improvement of inorganic aerosol component in PM2.5 by constraining aqueous-phase formation of sulfate in cloud with satellite retrievals: WRF-Chem simulations. *Sci. Total Environ.* 804, 150229 <https://doi.org/10.1016/j.scitotenv.2021.150229>.
- Sharma, E., Deo, R.C., Prasad, R., Parisi, A.V., 2020. A hybrid air quality early-warning framework: an hourly forecasting model with online sequential extreme learning machines and empirical mode decomposition algorithms. *Sci. Total Environ.* 709, 135934 <https://doi.org/10.1016/j.scitotenv.2019.135934>.
- Shin, U., Park, S.H., Park, J.S., Koo, J.H., Yoo, C., Kim, S., Lee, J.-Bum, 2021. Predictability of PM2.5 in Seoul based on atmospheric blocking forecasts using the NCEP global forecast system. *Atmos. Environ.* 246, 118141 <https://doi.org/10.1016/j.atmosenv.2020.118141>.
- Song, H., Deng, S.X., Lu, Z.Z., Li, J.H., Ba, L.M., Wang, J.F., Sun, Z.G., Li, G.H., Jiang, C., Hao, Y.Z., 2021. Scenario analysis of vehicular emission abatement procedures in Xi'an, China. *Environ. Pollut.* 269, 116187 <https://doi.org/10.1016/j.envpol.2020.116187>.
- Xu, M., Qin, Z., 2023. How does vehicle emission control policy affect air pollution emissions? Evidence from Hainan Province, China. *Sci. Total Environ.* 866, 161244 <https://doi.org/10.1016/j.scitotenv.2022.161244>.
- Zhang, H., Song, H., Wang, X., Wang, Y., Min, R., Qi, M., Ru, X., Bai, T., Xue, H., 2023. Effect of agricultural soil wind erosion on urban PM2.5 concentrations simulated by WRF-Chem and WEPS: a case study in Kaifeng, China. *Chemosphere* 323, 138250. <https://doi.org/10.1016/j.chemosphere.2023.138250>.
- Zhu, C., Qu, X., Qiu, M., Zhu, C., Wang, C., Wang, B., Sun, L., Yang, N., Yan, G., Xu, C., Li, L., 2023. High spatiotemporal resolution vehicular emission inventory in Beijing-Tianjin-Hebei and its surrounding areas (BTHSA) during 2000–2020, China. *Sci. Total Environ.* 873, 162389 <https://doi.org/10.1016/j.scitotenv.2023.162389>.